

* What is optimization: Optimization is a subject that is widely and increasingly used in science and technology, engineering, management, industry and on other areas. The basic goal of optimization is to solve any problem in very short calculation with an optimized solution. It is used in

Classification of optimization problem:

- i) Constrained optimization
- ii) Unconstrained optimization
- ii) Convex optimization

Feasible Direction: A vector $d \in \mathbb{R}^n$, $d \neq 0$, is called a feasible direction at $x \in \Omega$ if there exists $\alpha_0 > 0$ such that $x + \alpha d \in \Omega$ for all $\alpha \in [0, \alpha_0]$.

Local minimizer: A point $x^* \in \Omega$ is called local minimizer of f over Ω if there exists $\epsilon > 0$ such that $f(x) \geq f(x^*)$ for all $x \in \Omega$ and $\|x - x^*\| < \epsilon$.

Global minimizer: A point $x^* \in \Omega$ is a global minimizer of f over Ω if $f(x) \geq f(x^*)$ for $x \in \Omega$.

Graph: The graph of a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is the set of points in $\Omega \times \mathbb{R} \subset \mathbb{R}^{n+1}$.

Epigraph: The epigraph of a function, $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is the set of points in $\Omega \times \mathbb{R}$.

Convex function: A function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is called convex on Ω , if its epigraph is a convex set.

Convex set: A set $\Omega \subset \mathbb{R}^n$ is called convex if for every $x_1, x_2 \in \Omega$ and $\alpha \in (0, 1)$, $\alpha x_1 + (1-\alpha)x_2 \in \Omega$.

Derivative Matrix / Jacobian Matrix

Consider a differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

Let $f(x) = [f_1(x), f_2(x), \dots, f_m(x)]^T$

$$\therefore \frac{\partial f}{\partial x_j}(x_0) = \left[\frac{\partial f_1}{\partial x_j}(x_0), \frac{\partial f_2}{\partial x_j}(x_0), \dots, \frac{\partial f_m}{\partial x_j}(x_0) \right]^T$$

$$\therefore \left[\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0) \right] = \begin{bmatrix} \frac{\partial f_1(x_0)}{\partial x_1} & \dots & \frac{\partial f_1(x_0)}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m(x_0)}{\partial x_1} & \dots & \frac{\partial f_m(x_0)}{\partial x_n} \end{bmatrix}$$

This is called jacobian matrix.

Let $f(x_1, x_2) = 5x_1 + 8x_2 + x_1x_2 - x_1^2 - 2x_2^2$
find ∇f , Df , $F(x)$

Ans: $Df(x) = \begin{bmatrix} 5 + x_2 - 2x_1 & 8 + x_1 - 4x_2 \end{bmatrix}$

$$\therefore \nabla f = (Df(x))^T$$

$$= \begin{bmatrix} 5 + x_2 - 2x_1 \\ 8 + x_1 - 4x_2 \end{bmatrix}$$

$$F(x) = D^2 f(x)$$

$$= \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \\ 1 & -4 \end{bmatrix}$$

Feasible Direction:

* Define Feasible direction,

Ans: A vector $d \in \mathbb{R}^n$, $d \neq 0$, is called a feasible direction at $x \in \Omega$ if there exists such that $x + \alpha d \in \Omega$ for all $\alpha \in [0, \alpha_0]$.

* Define Directional derivative:

Ans: $\frac{\partial f(x)}{\partial d} = \nabla f(x)^T d = d^T \nabla f(x)$

* Define $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by $f(x) = x_1 x_2 x_3$ and Let $d = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right]$. Find $\frac{\partial f(x)}{\partial d}$

Ans: $f(x) = x_1 x_2 x_3$ and $d = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right]^T$

$$\therefore \nabla f(x) = \begin{bmatrix} x_2 x_3 \\ x_1 x_3 \\ x_1 x_2 \end{bmatrix}$$

The directional derivative of f in the direction d is

$$\frac{\partial f}{\partial d}(x) = \nabla f(x)^T d$$

$$= [x_2 x_3 \quad x_1 x_3 \quad x_1 x_2]$$

$$\begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix}$$

$$= \frac{x_2 x_3 + x_1 x_3 + \sqrt{2} x_1 x_2}{\sqrt{2}}$$

Prove

State and 1 First order necessary Condition:

Let Ω be a subset of $\mathbb{R}^n$ and $f \in C^1$ a real value function, and x^* is the minimizer

~~FON~~ Then $d^T \nabla f(x^*) \geq 0$.

Proof: We define

$$x(\alpha) = x^* + \alpha d \in \Omega$$

Here, $x(0) = x^*$ and Taylor's theorem,

$$\begin{aligned} f(x^* + \alpha d) - f(x^*) &= \phi(\alpha) - \phi(0) \\ &= \alpha d^T \nabla f(x(0)) + o(\alpha) \end{aligned}$$

where $\alpha \geq 0$, thus for sufficiently small value $\alpha > 0$, we have

$$d^T \nabla f(x^*) \geq 0$$

~~* State and~~

Consider the problem

$$\text{minimize } x_1^2 + 0.5x_2^2 + 3x_2 + 4.5$$

$$\text{subject to } x_1, x_2 \geq 0$$

Is the First order necessary condition (FONC) for a local minimizer satisfied

at) i) $x = [1, 3]^T$, (ii) $[0, 3]^T$, (iii) $[1, 0]^T$

iv) $[0, 0]^T$

Ans. Here, $\nabla f(x) = [2x_1, x_2 + 3]^T$

i) the point $x = [1, 3]^T$ does not satisfy FONC, because it requires $\nabla f(x) = 0$

(ii) $[0, 3]^T$ also doesn't satisfy FONC

(iii) $[1, 0]^T$ " " "

(iv) $[0, 0]^T$ satisfy FONC, $\nabla f(x) = 0$ Here.

Convex

Convex set and function

* Let $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subseteq \mathbb{R}^n$ is a convex set on Ω , then show that Ω is convex.

As contradiction proof
Proof: Suppose Ω is not convex set, then there exists y_1 and y_2 such that, $\alpha \in (0, 1)$
$$z = \alpha y_1 + (1 - \alpha) y_2 \in \Omega$$

Let $B_1 = f(y_1)$, $B_2 = f(y_2)$

Then the pairs $\begin{bmatrix} y_1 \\ B_1 \end{bmatrix}, \begin{bmatrix} y_2 \\ B_2 \end{bmatrix}$

$$\text{Let, } w = \alpha \begin{bmatrix} y_1 \\ B_1 \end{bmatrix} + (1 - \alpha) \begin{bmatrix} y_2 \\ B_2 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha y_1 + (1 - \alpha) y_2 \\ \alpha B_1 + (1 - \alpha) B_2 \end{bmatrix}$$

$$= \begin{bmatrix} z \\ \alpha B_1 + (1 - \alpha) B_2 \end{bmatrix}$$

then $z \notin \Omega$, $w \notin \text{epi}(f)$, Therefore f is not convex.

* Let $f(x) = x_1 x_2$, is f convex over

$$\Omega = \{x : x_1 \geq 0, x_2 \geq 0\}$$

Ans: Soln: Give

$$f(x) = x_1 x_2$$

Let $x = [1, 2]^T \in \Omega$ and $y = [2, 1]^T \in \Omega$

$$\text{Then } \alpha x + (1-\alpha)y = \alpha \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (1-\alpha) \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ = \begin{bmatrix} 2-\alpha \\ 1+\alpha \end{bmatrix}$$

$$\text{and } \alpha f(x) + (1-\alpha)f(y) = 2$$

~~$$\text{Let } \alpha = \frac{1}{2} \text{ (or)}$$~~

~~$$\text{Since } f(\alpha x + (1-\alpha)y)$$~~

$$\text{Since } f(\alpha x + (1-\alpha)y) > \alpha f(x) + (1-\alpha)f(y)$$

which shows that f is not convex over Ω .

Example 21.6:- Determine whether the following functions are convex, concave or neither

1. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$

2. $f: \mathbb{R}^3 \rightarrow \mathbb{R}, f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$

3. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$

Solution:-

1. Given that

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$$

$$\text{Now } \nabla f(x) = -16x$$

The Hessian matrix of f is

$$F(x) = -16 < 0, \text{ for all } x \in \mathbb{R}.$$

Hence, f is concave over $\mathbb{R}$.

2. Given that

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$$

$$\text{Now } \nabla f(x) = \begin{bmatrix} 8x_1 + 6x_2 + x_3 - 3 \\ 6x_2 + 6x_1 - 2 \\ 10x_3 + x_1 \end{bmatrix}$$

The Hessian matrix of f is

$$F(x) = \begin{bmatrix} 8 & 6 & 1 \\ 6 & 6 & 0 \\ 1 & 0 & 10 \end{bmatrix}$$

The leading principal minors of $F(x)$ are

$$\Delta_1 = 8 > 0$$

$$\Delta_2 = \det \begin{bmatrix} 8 & 6 \\ 6 & 6 \end{bmatrix} = 12 > 0$$

$$\Delta_3 = \det F(x) = 114 > 0$$

Hence, $F(x)$ is positive definite for all $x \in \mathbb{R}^3$.
Therefore, f is a convex over $\mathbb{R}^3$.

3. Given that

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$$

$$\text{Now } \nabla f(x) = \begin{bmatrix} 2x_2 - 2x_1 \\ 2x_1 - 2x_2 \end{bmatrix}$$

The Hessian of f is

$$F(x) = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 0$$

which is negative semidefinite for all $x \in \mathbb{R}^2$.
Hence f is concave on $\mathbb{R}^2$.

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